Market Crashes as Phase Transitions: Simulating Market Crashes with the Ising Model

Joven Sekhon

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1 Introduction

Traditional financial models often assume that price movements follow stochastic processes such as Geometric Brownian Motion (GBM), where returns are independent and volatility is constant. Although this framework forms the basis for many models in quantitative finance, including Black–Scholes, it struggles to capture several well-documented empirical features of financial markets. In reality, markets display phenomena such as volatility clustering (where large price changes are followed by large changes, and vice versa), herding behaviour among traders, and sudden market crashes.[3, 6]

Interestingly, similar types of collective behaviour arise in statistical physics. One of the simplest models capturing this phenomenon is the Ising model, originally developed to study magnetisation in ferromagnetic materials.This article explores how ideas from statistical physics can be used to model financial markets. By interpreting trader decisions as interacting agents within an Ising-type system, it becomes possible to study how local interactions between traders can lead to these large-scale collective dynamics evident in financial markets. Using Monte Carlo simulation methods, the article will demonstrate how varying interaction strength and randomness in such models can generate behaviours that resemble real market dynamics.

2 Limitations of Classical Finance Models

Geometric Brownian motion (GBM) is a widely used model for asset prices, forming the basis of Black–Scholes. It assumes that price changes are driven by a combination of deterministic drift and stochastic fluctuations modelled as Brownian motion by the stochastic differential equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{1}$$

Taking logarithms, returns follow:

$$d\log S_t = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t, \tag{2}$$

implying that returns are normally distributed and independent over time.

However, empirical return distributions exhibit heavy tails; large market crashes occur more abruptly and with greater magnitude than can be explained by Gaussian fluctuations alone [6, 3].

Furthermore, GBM assumes independence between price changes. However, real markets display strong correlations arising from collective behaviour and herding effects among agents.[3, 2] These empirical observations highlight a fundamental limitation of classical models: they assume independent, small fluctuations and fail to account for interactions between agents that can give rise to large, collective movements.

This suggests that alternative modelling approaches are required. In this context, models from statistical physics, such as the Ising model, provide a natural framework for studying how local interactions can lead to large-scale market dynamics.

3 The Ising Model In Statistical Physics

The Ising Model, invented by German physicist Wilhelm Lenz in 1920 as a model for ferromagnetism, is now a foundational tool in statistical mechanics for understanding phase transitions. In this Article, we will only consider the two-dimensional square-lattice Ising model. [4]

3.1 Model Setup

The (2-D) Ising model consists of a lattice of N sites, each associated with a spin variable $s_i \in \{-1, 1\}$. Interactions occur between neighbouring spins; in 2-D defined on a two-dimensional square lattice with periodic boundary conditions.

3.2 The Hamiltonian

The behaviour of the system is governed by the Hamiltonian, H, which represents as the total energy of a given spin configuration. It is defined as:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum s_i \tag{3}$$

where $s_i \in \{-1, 1\}$ represents the spin at site i , J is the interaction strength between neighbouring spins, h represents an external field, and $\langle \cdot \rangle$ indicates to only sum over adjacent spins. The first term captures interactions between neighbouring spins. When $J > 0$ aligned neighbour spins ($s_i = s_j$) reduce the energy of the system, encouraging local ordering. Conversely, misaligned spins increase the energy, making such configurations less favourable. The second term represents the influence of an external field, which biases spins towards a particular orientation.

3.3 Boltzmann Distribution and Temperature

The probability of a given spin configuration $\{s_i\}$ is governed by the Boltzmann distribution:

$$P(\{s_i\}) = \frac{1}{Z} e^{-\beta H} \tag{4}$$

where $\beta = \frac{1}{k_B T}$, T is the temperature, and Z is the partition function given by

$$Z = \sum_{\{s_i\}} e^{-\beta H}. \tag{5}$$

This distribution assigns higher probability to configurations with lower energy, while still allowing higher-energy states to occur with non-zero probability. At high temperatures ($T \to \infty$), β becomes small, and the exponential weighting becomes less significant. Therefore, configurations of all energies occur with similar probability, leading to disordered states. At low temperatures ($T \to 0$), β becomes large, and the exponential term suppresses high-energy configurations. As a result, the system settles into states that minimises the Hamiltonian, leading to ordered configurations. Thus, temperature acts as a parameter governing the balance between order and randomness determining whether local interactions or stochastic fluctuations dominate.

3.4 Magnetisation

A key Macroscopic observable in the Ising model is the magnetisation, defined as

$$M = \sum_i s_i \quad m = \frac{1}{N} \sum_i s_i \tag{6}$$

where $s_i \in \{-1, 1\}$ and N is the total number of spins. The quantity m represents the average magnetisation per spin

The magnetisation measures the degree of alignment in the system. At lower temperatures, spins tend to align and the system is in an ordered state. Here, $m \approx \pm 1$. When $m \approx 0$, the system is disordered, with no dominant orientation. This tends to occur at higher temperatures.

3.5 Phase Transitions and Critical Temperature

As the temperature varies, the Ising model undergoes a phase transition between ordered and disordered states. This transition occurs at a critical temperature T_c ($T = 2.269$ in 2D), which marks a qualitative change in the system's behaviour. Near the critical temperature, the system displays large-scale fluctuations, increased variance in magnetisation, and sensitivity to small perturbations. These features are reflected in peaks of quantities such as susceptibility and specific heat. This critical regime allows local interactions to propagate across the entire system, leading to collective behaviour not seen at higher or lower temperatures.

4 Mapping the Ising Model to Financial Markets

The key idea is to interpret the abstract components of the Ising model as representing elements of a financial market. While the mapping is not exact, it provides a useful conceptual framework for modelling collective behaviour.

4.1 Agents as Spins

In this framework, each $s_i \in \{-1, 1\}$ is interpreted as an individual trader or agent in the market.The two possible states correspond to binary trading decisions:

$$s_i = \begin{cases} +1 & \text{buy (bullish)} \\ -1 & \text{sell (bearish)} \end{cases} \tag{7}$$

This simplification ignores position size and more complex strategies, but captures the essential directional behaviour of market participants.

4.2 Interactions and Herding Behaviour

The interaction term in the Hamiltonian:

$$-J \sum_{\langle i,j \rangle} s_i s_j$$

can be interpreted as modelling social influence or herding effects between traders.[2, 9] When $J > 0$, neighbouring agents have an incentive to align their decisions. This reflects mechanisms in financial markets like imitation of other traders' behaviour, information cascades and momentum-driven strategy. This is precicely what classical models like GBM fail to capture: returns are not independent because decisions are not independent. This introduces correlations between agents, meaning that local interactions can propagate through the system and generate large-scale collective behaviour.

4.3 Magnetisation as Market Sentiment

Magnetisation provides a natural aggregate quantity:

$$m = \frac{1}{N} \sum_i s_i$$

In this framework, m represents market sentiment or order imbalance. $m \approx \pm 1$ represent strong buying/selling pressures, and $m \approx 0$ represent a balanced market. Fluctuations in magnetisation therefore capture changes in aggregate market behaviour over time, providing a natural link between micro-level interactions and macro-level dynamics.

4.4 External Field as Market Bias

The external field term:

$$-h \sum s_i$$

introduces a global bias affecting all agents simultaneously. In financial contexts, this can represent macroeconomic news, central bank announcements or geopolitical events. A positive h biases agents towards buying, whilst a negative one biases towards selling.

To incorporate more realistic market dynamics, we will also extend the standard Ising framework using the Bornholdt model. In this formulation, an additional macro field acting on each agent is given by:

$$-\alpha s_i \left| \frac{1}{N} \sum_j s_j \right| \tag{8}$$

where this term introduces a global feedback mechanism proportional to average magnetisation:[1]

The parameter $\alpha > 0$ controls the strength of this feedback. When the market is strongly aligned (i.e. $|m(t)|$ is large), the second term penalises agents for following the majority, thereby stabilising minority positions. This creates a competition between local imitation and global contrarian behaviour, leading to intermittent dynamics and providing a mechanism through which features such as volatility clustering can emerge.[1]

4.5 Temperature as Market Uncertainty

Temperature, T, plays a central role in the mapping. In financial markets, it can be interpreted as a measure of noise or uncertainty.

At high temperatures, thermal noise dominates interactions, and agents behave approximately independently. In this regime, aggregate behaviour resembles classical models such as GBM, where price changes are largely uncorrelated. At low temperatures, interactions dominate, leading to strong alignment between agents and persistent collective behaviour. Near the critical temperature, the system exhibits heightened sensitivity to small perturbations, with local changes capable of triggering large-scale shifts in the global state. A precise financial analogue of temperature is not directly observable, so in this model T should be interpreted as an effective parameter rather than a literal market variable. Quantitatively, it controls the balance between idiosyncratic randomness in individual decisions and the tendency of agents to align through interaction. In practice, one could estimate an effective temperature by calibrating the model so that simulated time series reproduce empirical market properties such as return variance, kurtosis, or volatility autocorrelation.

4.6 From Magnetisation to Returns

While magnetisation provides a measure of aggregate alignment in the system, it is not directly equivalent to financial returns, hence it is necessary to construct a synthetic return series from the dynamics of magnetisation.

We define returns as the change in magnetisation between successive time steps:

$$r_t = m_{t+1} - m_t \tag{9}$$

This reflects the idea that price changes arise from shifts in aggregate market sentiment; a large number of agents simultaneously changing their state leads to a large shift in magnetisation. Conversely, small or uncoordinated changes produce only minor fluctuations. In particular, near the critical temperature, large coordinated shifts in magnetisation can occur, leading to heavy-tailed return distributions, providing a potential mechanism for the emergence of extreme market events within the model.

5 Simulating Market Dynamics with Monte Carlo Simulations

This mapping suggests that large market movements need not be driven by external shocks alone, but can instead emerge endogenously from the interaction of agents within the system. To investigate these dynamics, we simulate the evolution of the system using Monte Carlo methods, specifically a Markov Chain Monte Carlo using the Metropolis algorithm.

5.1 Simulation Setup

The system is defined on a two-dimensional square lattice of size $L \times L$ with periodic boundary conditions.

Periodic boundary conditions eliminate edge effects by effectively wrapping the lattice onto a torus. This ensures that all spins have the same number of neighbours, allowing the system to better approximate an infinite lattice and preventing artificial boundary artifacts from influencing the dynamics. Each site is assigned a spin $s_i \in \{-1, 1\}$, initialised randomly.

5.2 Metropolis Algorithm

To simulate the evolution of the system, we use the Metropolis algorithm. At each step, a spin s_i is selected and a flip $s_i \to -s_i$ is proposed [7]. The resulting energy change can be derived from the Hamiltonian

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i.$$

Only the terms involving s_i are affected by the flip, so the relevant local contribution is

$$H_i = -J \sum_{j \in \mathrm{nn}(i)} s_i s_j - h s_i.$$

After the flip, this becomes

$$H_i' = -J \sum_{j \in \mathrm{nn}(i)} (-s_i) s_j - h(-s_i),$$

and hence the change in energy is

$$\Delta E = H_i' - H_i = 2s_i \left(J \sum_{j \in \mathrm{nn}(i)} s_j + h \right).$$

This expression shows that the cost of flipping a spin depends only on its local neighbourhood and the external field, speeding up computation as the Hamiltonian does not have to be recomputed for each proposed spin alignments. The proposed flip is then accepted with probability:

$$P(\text{accept}) = \begin{cases} 1, & \Delta E \leq 0, \\ e^{-\beta \Delta E}, & \Delta E > 0. \end{cases}$$

This rule favours energetically favourable flips (reducing the Hamiltonian) while still allowing higher-energy moves with non-zero probability. When implementing the Bornholdt update rule, the energy change takes the same form as in the standard Ising model, but with the effective local field modified to include the global feedback term:

$$\Delta E = 2s_i \left(J \sum_{j \in \mathrm{nn}(i)} s_j + h - \alpha s_i |m| \right),$$

where the derivation follows identically from substituting the modified local field into the standard expression.

5.3 Measured Quantities

During the simulation, we record a set of key observables that characterise the state of the system and allow us to connect the model to both statistical physics and financial interpretations. At each Monte Carlo step t, we compute the magnetisation and also compute the energy of the system,

From these time series, we estimate ensemble-averaged quantities [**?** 8]. In particular, we compute the susceptibility,

$$\chi = \frac{N}{T} \left(\langle m^2 \rangle - \langle m \rangle^2 \right), \tag{10}$$

which measures the variance of magnetisation and captures the system's sensitivity to fluctuations. We also compute the specific heat:

$$C_v = \frac{1}{NT^2} \left(\langle E^2 \rangle - \langle E \rangle^2 \right), \tag{11}$$

which reflects fluctuations in energy and provides another indicator of critical behaviour. These quantities are particularly important near the critical temperature, where both χ and C_v exhibit peaks, signalling large-scale collective dynamics [**?**]. In this simulation, we define and record the time series of returns defined as:

$$r_t = m_{t+1} - m_t \tag{12}$$

This allows the statistical properties of the simulated system to be analysed using standard tools from quantitative finance. In addition to returns, we also consider derived quantities such as absolute returns and rolling volatility; The absolute return is defined as:

$$|r_t| = |m_{t+1} - m_t|\,, \tag{13}$$

which serves as a simple proxy for market volatility [3]. To capture time-varying volatility, we define the rolling volatility over a window of size w as:

$$\sigma_t = \sqrt{\frac{1}{w}\sum_{k=0}^{w-1}(r_{t-k} - \bar{r}_t)^2}, \tag{14}$$

where $\bar{r}_t$ is the rolling mean return over the same window,

$$\bar{r}_t = \frac{1}{w}\sum_{k=0}^{w-1} r_{t-k}. \tag{15}$$

To quantify temporal dependence in the simulated market series and identify volatility clustering, we compute the autocorrelation function (ACF) of returns and of both block and rolling volatility. The ACF is the covariance of a time series with a lagged version of itself, normalised by its variance, given by the formula:

$$\rho_x(\tau) = \frac{\mathbb{E}\big[(x_t - \mu)(x_{t+\tau} - \mu)\big]}{\mathbb{E}\big[(x_t - \mu)^2\big]} \tag{16}$$

where $\mu = \mathbb{E}[x_t]$ is the mean of the time series. The autocorrelation function measures how strongly values of the series are related across time lags, allowing us to detect persistence or memory in returns and volatility. We apply this to r_t, $|r_t|$, and volatility measures.

5.4 Simulation Parameters

The simulations are performed on a two-dimensional square lattice of size $L \times L$, with periodic boundary conditions, giving a total of $N = L^2$ spins. We consider a range of temperatures:

$$T \in \{1.0, 1.5, 2.0, 2.269, 2.5, 3.0, 4.0, 6.5\},$$

The temperatures used in the simulations were chosen to sample the three qualitatively distinct regimes of the Ising system: an ordered regime ($T < T_c$), a critical regime

($T \approx T_c$), and a disordered regime ($T > T_c$). The aim is therefore not to claim that a real market literally has temperature $T = 2.269$, but rather to examine how market-like observables change as the effective balance between coordination and randomness is varied across regimes.

For each temperature, the system is evolved using the Metropolis algorithm updated by Monte Carlo Sweeps. Each sweep consists of N attempted spin updates, ensuring that, on average, every spin is updated once per sweep.

5.5 Implementation Difficulties

Several practical difficulties arose in the simulations. Near the critical temperature, the system exhibits critical slowing down, meaning that successive configurations remain correlated for extended periods. This led to significantly longer simulations (order of 10^6 Monte Carlo sweeps) to obtain reliable estimates of equilibrium quantities.

In practice, each run at a fixed temperature and α required approximately 40 minutes of computation time. This imposes a trade-off between statistical accuracy and computational cost. For larger-scale studies or finer parameter sweeps, the use of performance optimisations such as just-in-time compilation (e.g. via `numba`) or parallelisation would be highly beneficial.

Despite these longer runs, the finite lattice size remains a limiting factor. In particular, the standard Ising model does not generate the long-range temporal dependence in volatility observed in real financial markets. While large fluctuations and heavy-tailed returns emerge near criticality, the autocorrelation of volatility decays relatively quickly. This limitation motivates the use of extensions such as the Bornholdt model, which introduce additional feedback mechanisms capable of producing more persistent volatility dynamics.

The code used to generate all simulations and figures is available in a public GitHub repository, allowing the results to be reproduced and extended - `https://github.com/JSek7/Ising-Model-Market-Crash-Simulation`

6 Results

6.1 Thermodynamic Behaviour

We first analyse the behaviour of standard thermodynamic observables as a function of temperature.

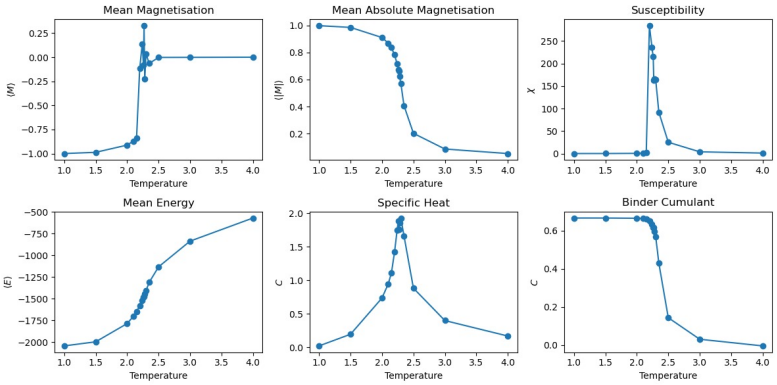


Figure 1: Thermodynamic observables as a function of temperature.

At low temperatures ($T < T_c$), the system exhibits strong ordering, with the average magnetisation satisfying $|m| \approx 1$. This corresponds to a state in which most agents adopt the same trading decision, leading to highly coordinated market behaviour. In financial terms, this regime resembles strongly trending markets driven by herding, where price movements are persistent and returns are relatively stable in direction but small in magnitude due to limited changes in aggregate sentiment.

As the temperature increases, the magnetisation decreases towards zero, reflecting a transition to a disordered state in which agents act more independently. This regime is analogous to highly liquid or information-efficient markets, where individual actions are less correlated and price movements resemble noise. Here, returns are less predictable and exhibit higher variability, but without strong collective structure.

Around the critical temperature $T_c \approx 2.269$, the system undergoes a sharp transition between these regimes. In this critical region, the system displays large fluctuations in magnetisation, corresponding to rapid shifts in aggregate market sentiment. These fluctuations translate into large changes in the synthetic return series, producing heavy-tailed behaviour and an increased likelihood of extreme events.

This transition is further characterised by peaks in both the susceptibility χ and specific heat C_v. The susceptibility, which measures fluctuations in magnetisation, can be interpreted as a proxy for market sensitivity: near criticality, small perturbations—such as minor news events or local changes in trader behaviour—can trigger disproportionately large market responses. Similarly, the peak in specific heat reflects large energy fluctuations, corresponding to heightened instability in the system.

6.2 Magnetisation Time Series

The time series of magnetisation provides further insight into the dynamics of the system.

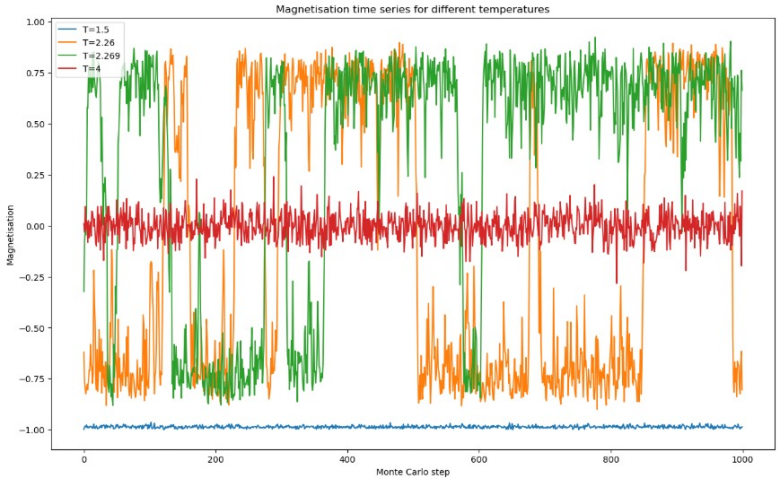


Figure 2: Magnetisation Time Series for T = 1.5, 2.26, 2.269, 4

At low temperatures, the system rapidly settles into one of the two ordered states ($m \approx \pm 1$) and remains there, with only small fluctuations. This reflects a stable, highly coordinated regime.

At high temperatures, the magnetisation fluctuates randomly around zero, indicating a disordered system with little correlation between spins.

Near the critical temperature, the system exhibits large, persistent fluctuations, with periods of relative stability interrupted by sudden shifts in magnetisation. This regime is characterised by long-range correlations and increased variability.

6.3 Synthetic Returns

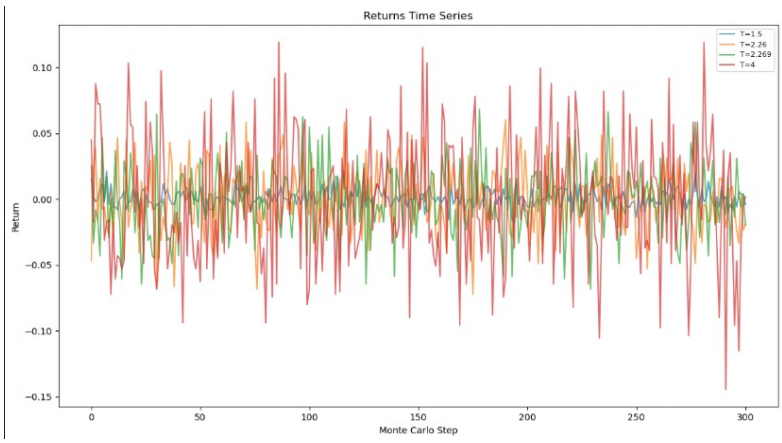
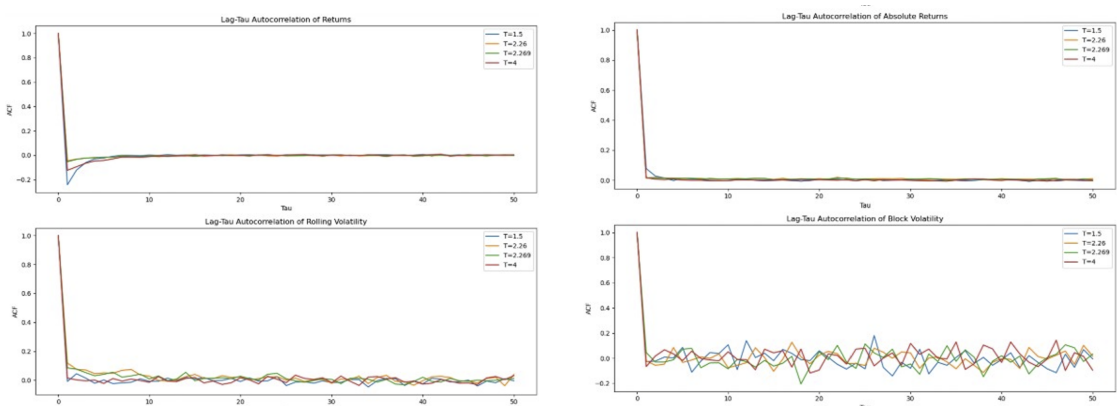


Figure 3: Returns Time Series for T = 1.5, 2.26, 2.269, 4

At low temperatures, returns are small, reflecting the stability of the ordered state. At high temperatures, returns are larger and resemble noise, consistent with a system dominated by randomness. Near the critical temperature, the return series exhibits heavy tails, with more frequent large fluctuations than would be expected under a Gaussian model. [6, 3] This behaviour is qualitatively consistent with empirical financial data.

6.4 Volatility and Autocorrelation

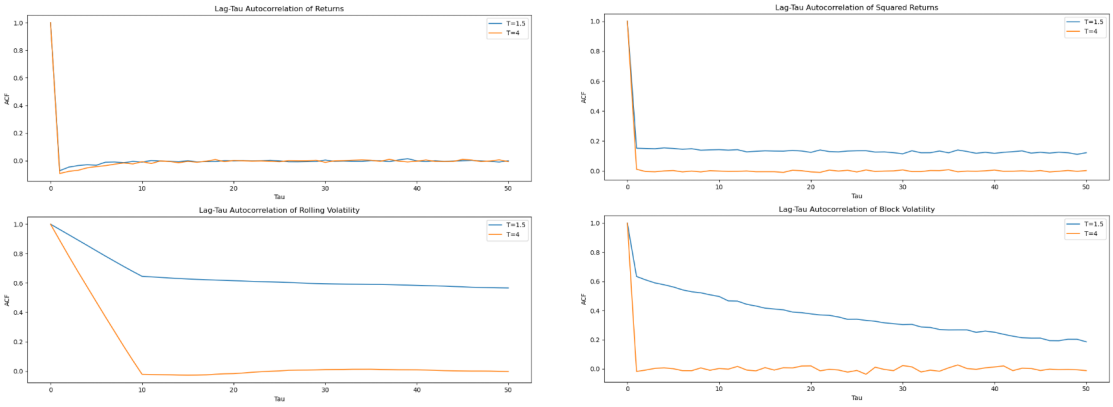
To investigate temporal dependencies, we analyse the autocorrelation functions of returns and volatility proxies.



(a) Returns and Rolling Volatility (b) Squared Returns and Block Volatility

Figure 4: Auto-Correlation Functions for Different Temperatures

The autocorrelation of returns rapidly decays to zero, indicating little linear predictability, consistent with real financial markets. The autocorrelation of squared returns, rolling volatility and block volatility also decays quickly, confirming that the standard Ising model fails to reproduce volatility clustering. Using the Bornholdt Update Rule however, we can see that this is not the case:



(a) Returns and Rolling Volatility

(b) Squared Returns and Block Volatility

Figure 5: Auto-Correlation Functions with Bornholdt Rule for Different Temperatures

While raw returns remain largely uncorrelated, the autocorrelation of squared returns and volatility decays much more slowly under the Bornholdt dynamics. This indicates the presence of memory in the magnitude of fluctuations, even though the direction of returns remains unpredictable.[1, 5]

7 Conclusion and Limitations

One limitation is defining returns as $r_t = m_{t+1} - m_t$, imposing a linear relationship between aggregate sentiment and price changes. In real markets, price formation is more complex and may involve nonlinear responses, delayed effects, and feedback mechanisms. Alternative constructions of returns, for example incorporating multiplicative noise or nonlinear transformations of magnetisation, could potentially produce more realistic volatility behaviour.

Another important simplification is the assumption of homogeneous agents and static interactions. In the standard Ising model, all spins interact with equal strength and follow identical update rules. In contrast, financial markets consist of heterogeneous agents with differing time horizons, strategies, and information sets. Introducing heterogeneity or adaptive interactions could allow the model to capture more realistic features such as herding behaviour, regime shifts, and endogenous changes in market volatility.

Several extensions could address these shortcomings. Allowing the temperature to vary over time would introduce a mechanism for changing market uncertainty, whilst more sophisticated variants, such as agent-based models, may also generate longer memory and improved volatility clustering.

Overall, while the Ising model provides a powerful conceptual framework for understanding how local interactions can lead to global market phenomena, it should be viewed as a starting point rather than a complete description of financial dynamics. Future work should focus on relaxing these assumptions to bridge the gap between stylised physical models and empirical financial data.

This article has explored how the Ising model can be used to study collective behaviour in financial markets. By modelling traders as interacting agents, the framework provides a natural mechanism for generating correlated behaviour and large-scale market movements.

The Standard Ising model and Bornholdt Rule successfully captures several key qualitative features relevant to financial systems:

- Collective behaviour emerging from local interactions
- Large fluctuations near a critical point
- Heavy-tailed return distributions arising endogenously
- Volatility Clustering (with the Bornholdt Rule only)

In particular, the critical regime ($T \approx T_c$) plays a central role. Near criticality, small local changes can propagate throughout the system, leading to large, system-wide fluctuations. This provides an interesting interpretation of market instability, where crashes may arise endogenously from the amplification of small local perturbations near criticality.

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